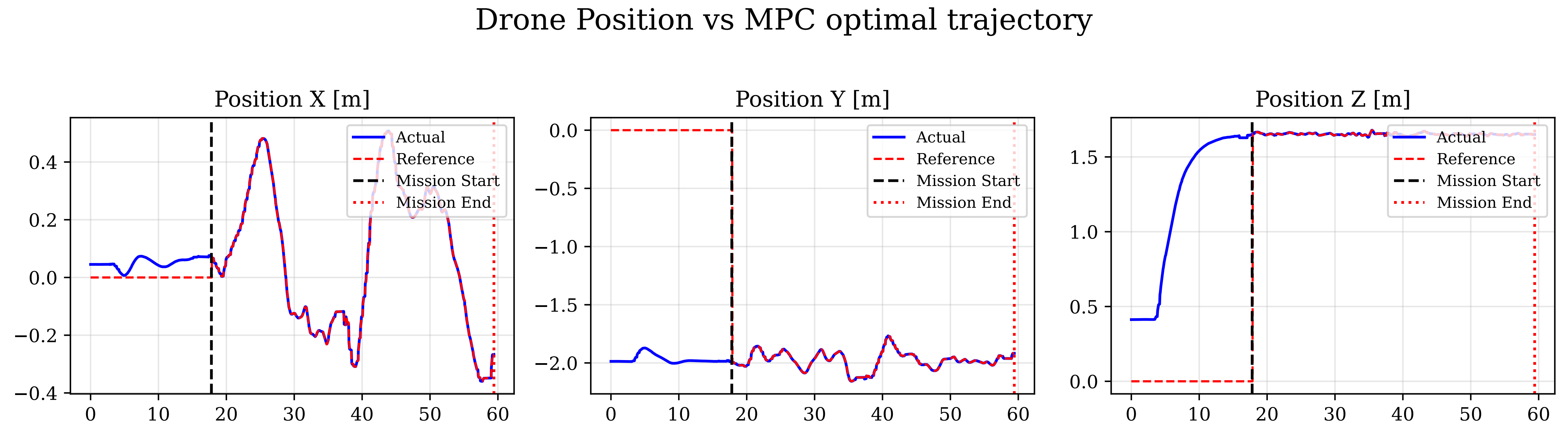
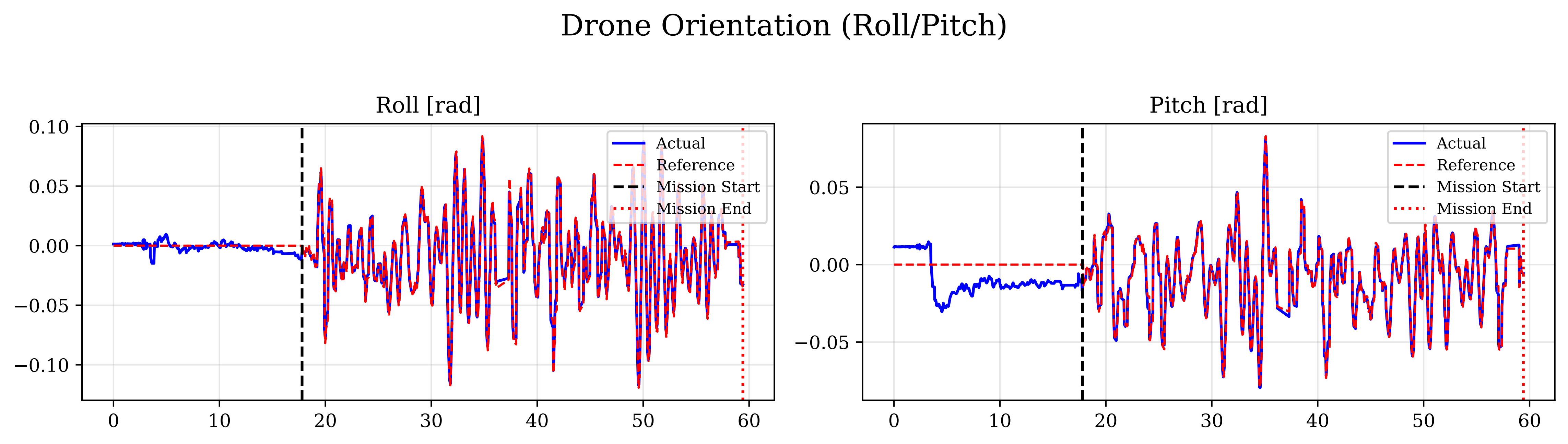
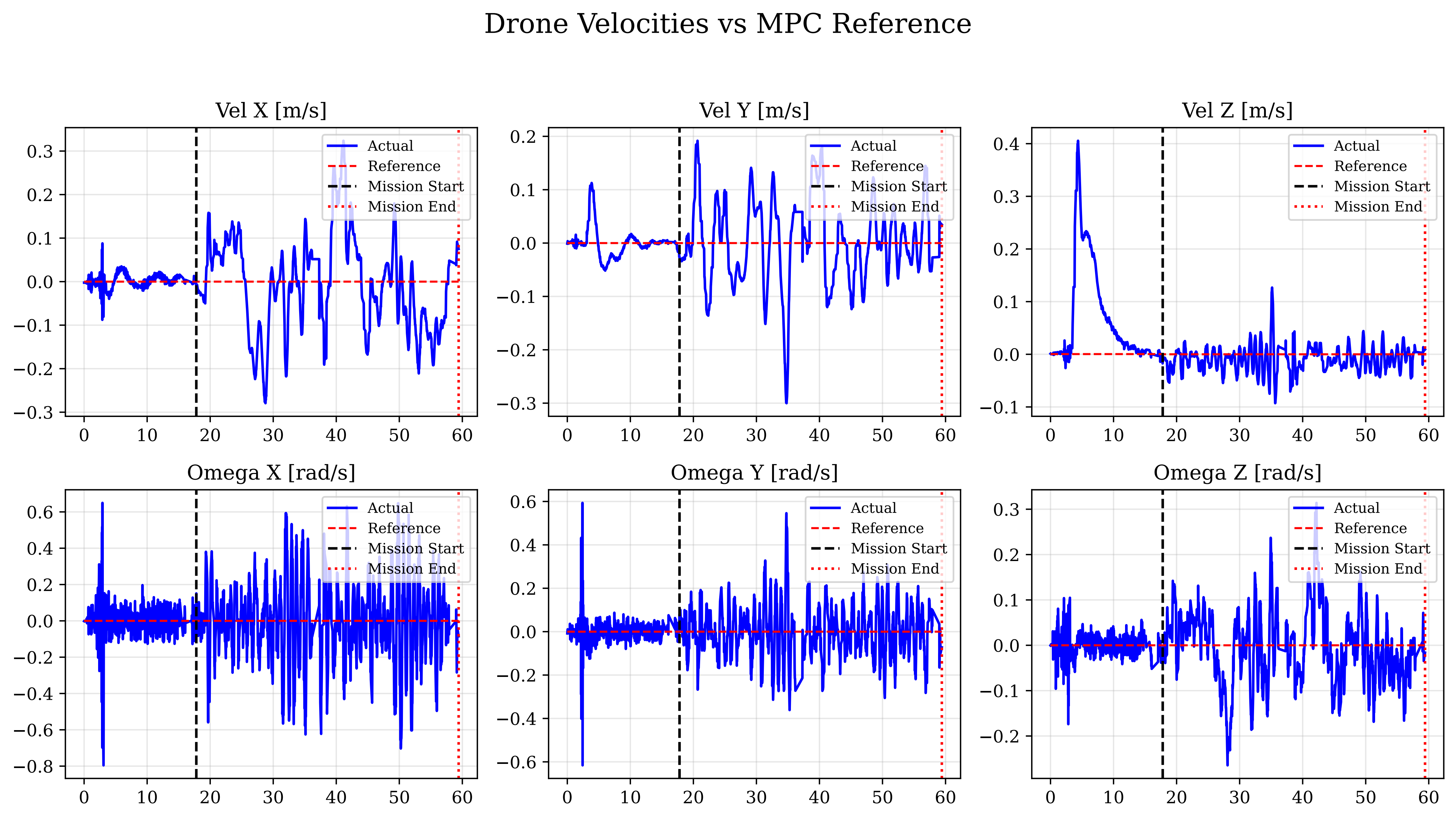
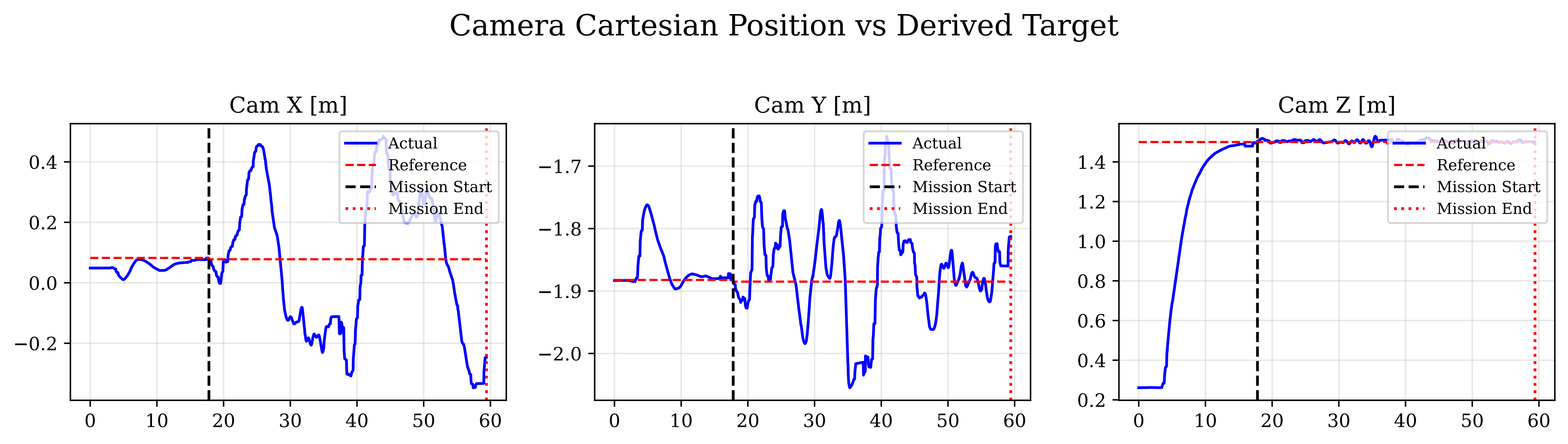


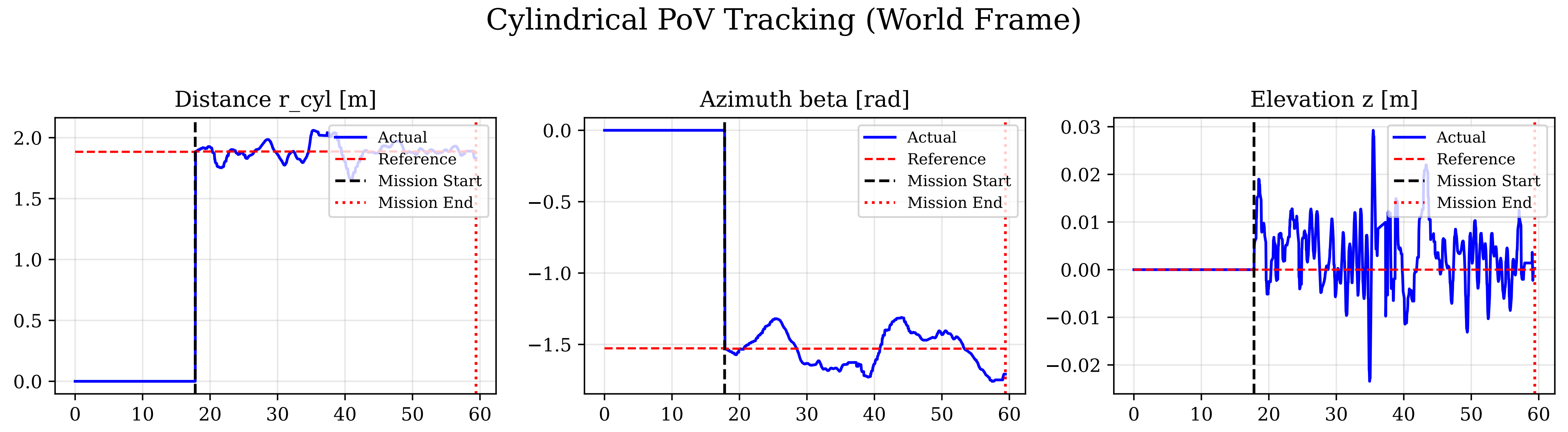
Drone Position vs MPC optimal trajectory
Position X [m]
Position Y [m]
Position Z [m]
Actual
Reference
Mission Start
Mission End


Drone Orientation (Roll/Pitch)
Roll [rad]
Pitch [rad]
Actual
Reference
Mission Start
Mission End
0.10
0.05
0.00
−0.05
−0.10
0
10
20
30
40
50
60

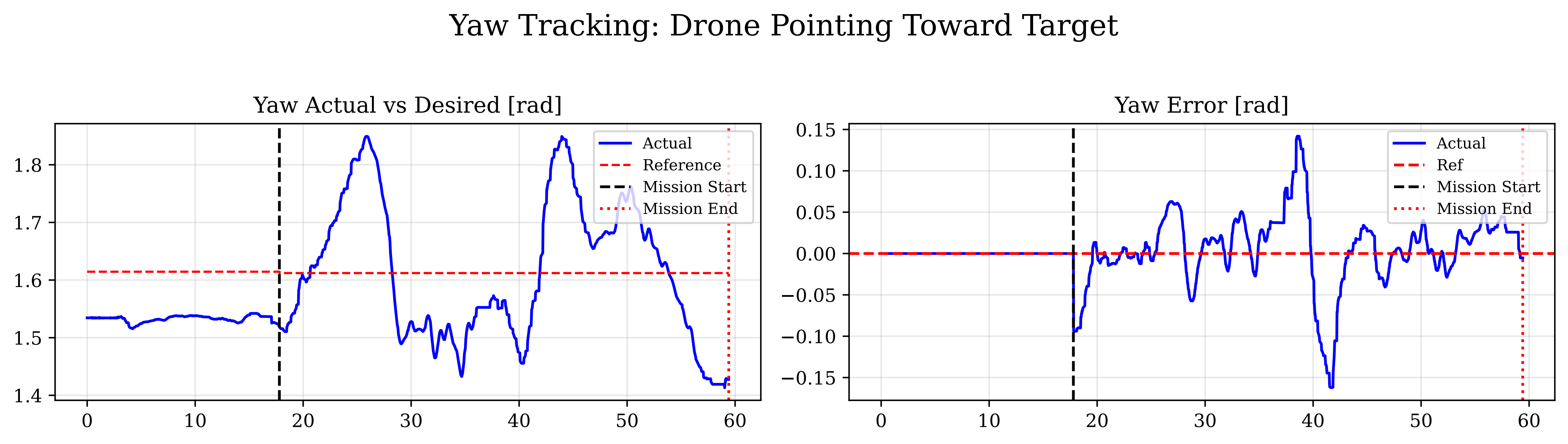


Drone Velocities vs MPC Reference
Vel X [m/s]
Vel Y [m/s]
Vel Z [m/s]
Omega X [rad/s]
Omega Y [rad/s]
Omega Z [rad/s]
Actual
Reference
Mission Start
Mission End

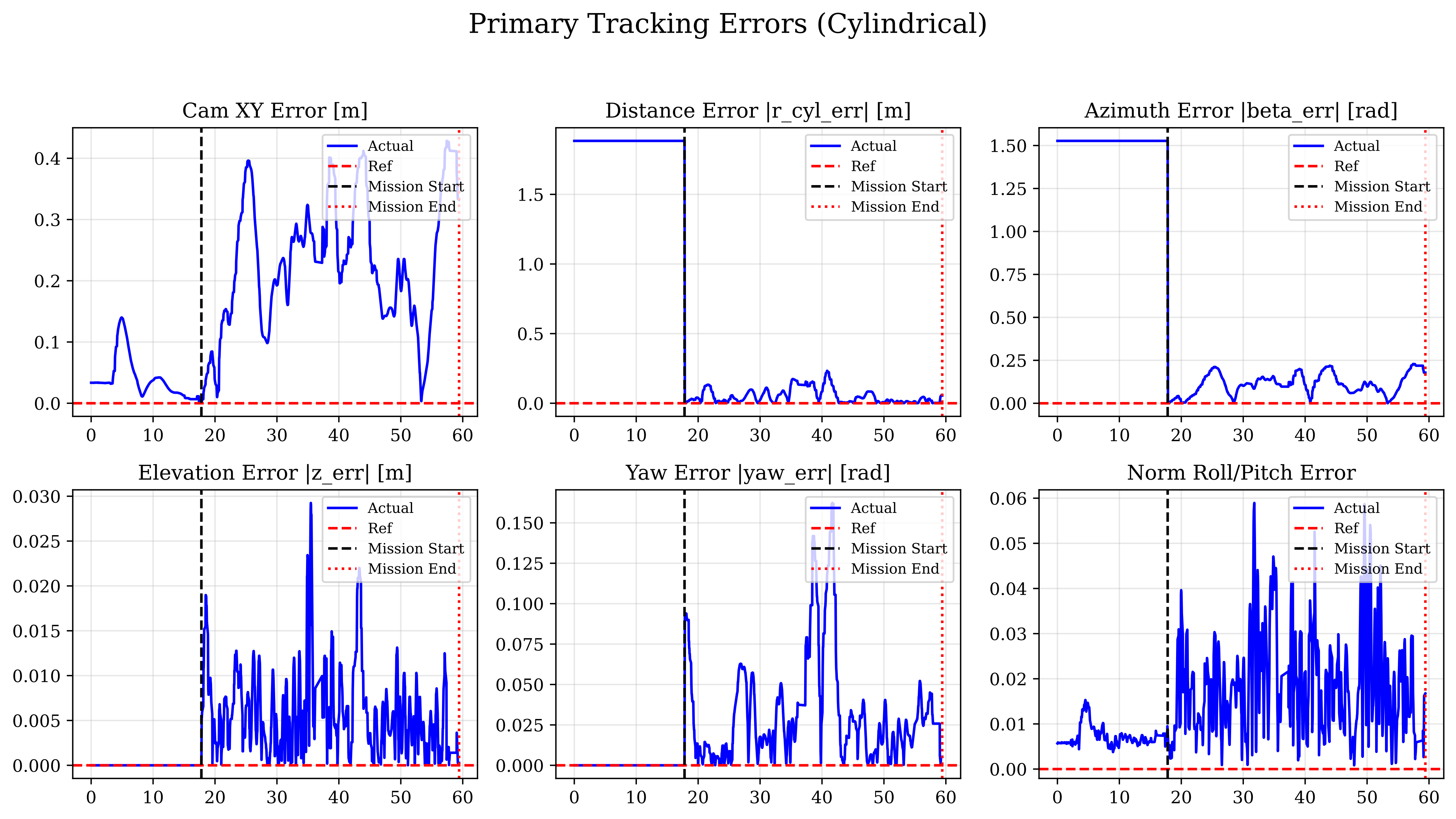

Camera Cartesian Position vs Derived Target
Cam X [m]
Cam Y [m]
Cam Z [m]
Actual
Reference
Mission Start
Mission End
0.4
0.2
0.0
−0.2
−1.7
−1.8
−1.9
−2.0
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0
10
20
30
40
50
60

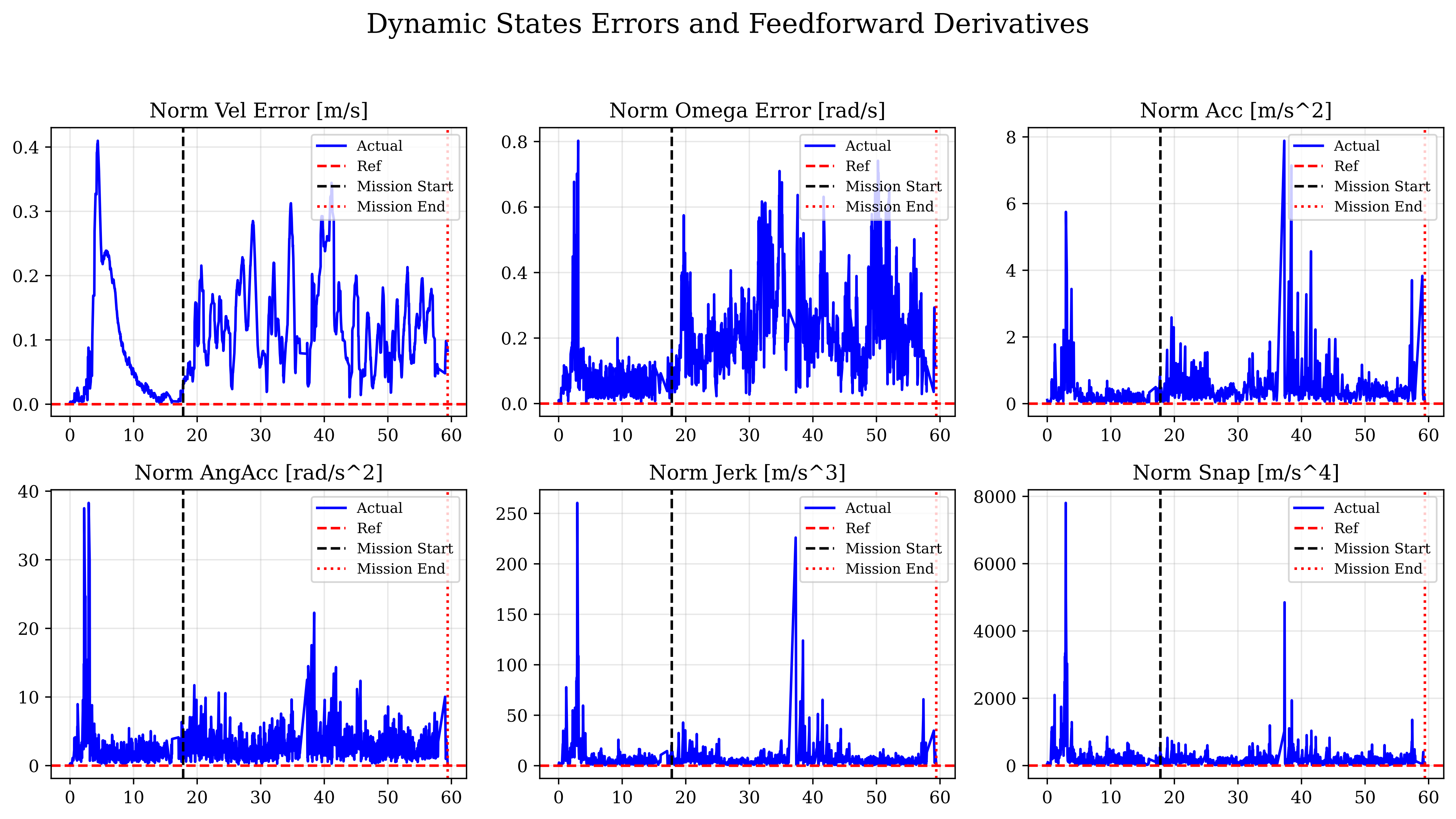


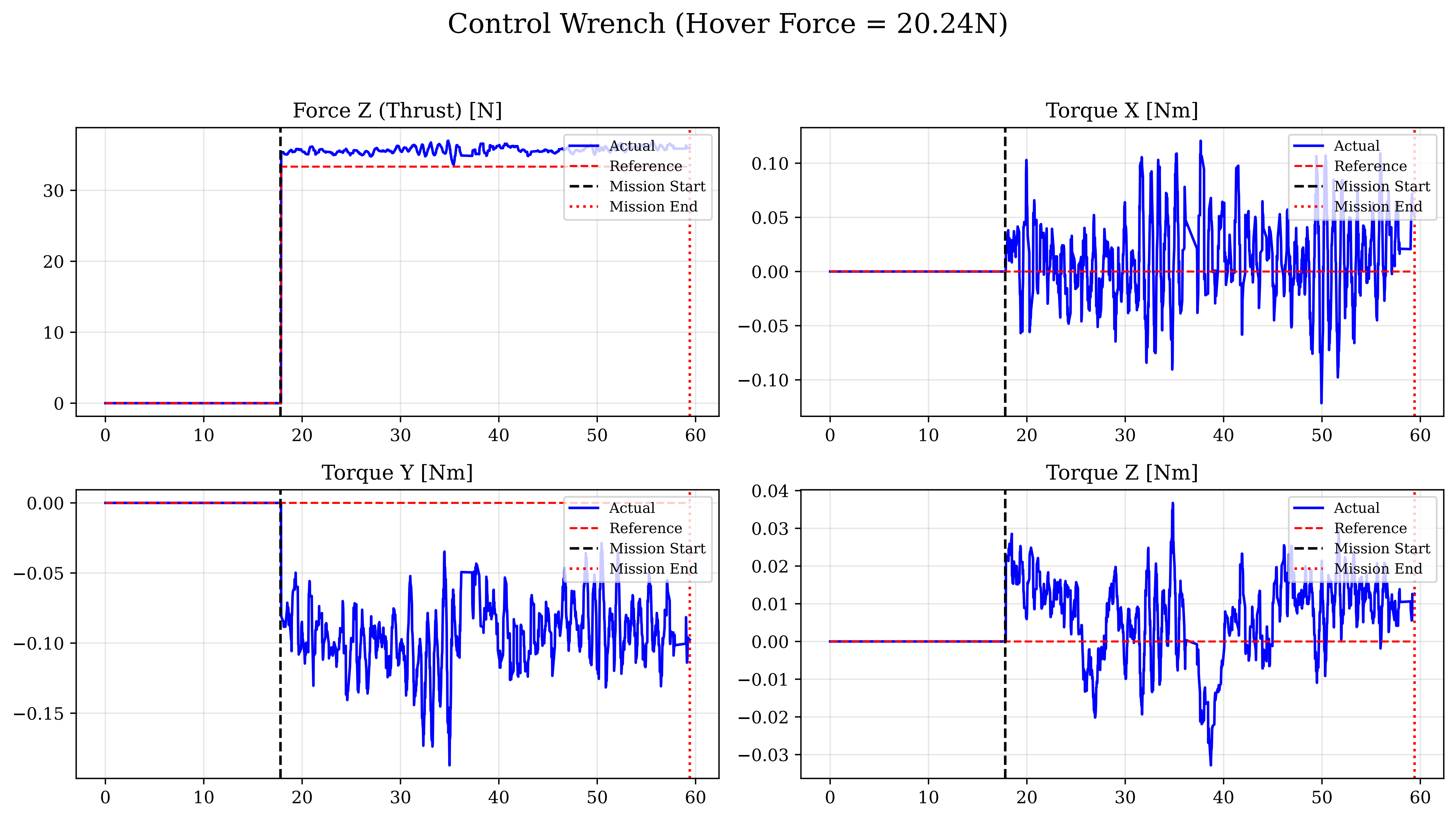
Cylindrical PoV Tracking (World Frame)
Distance r_cyl [m]
Azimuth beta [rad]
Elevation z [m]
Actual
Reference
Mission Start
Mission End



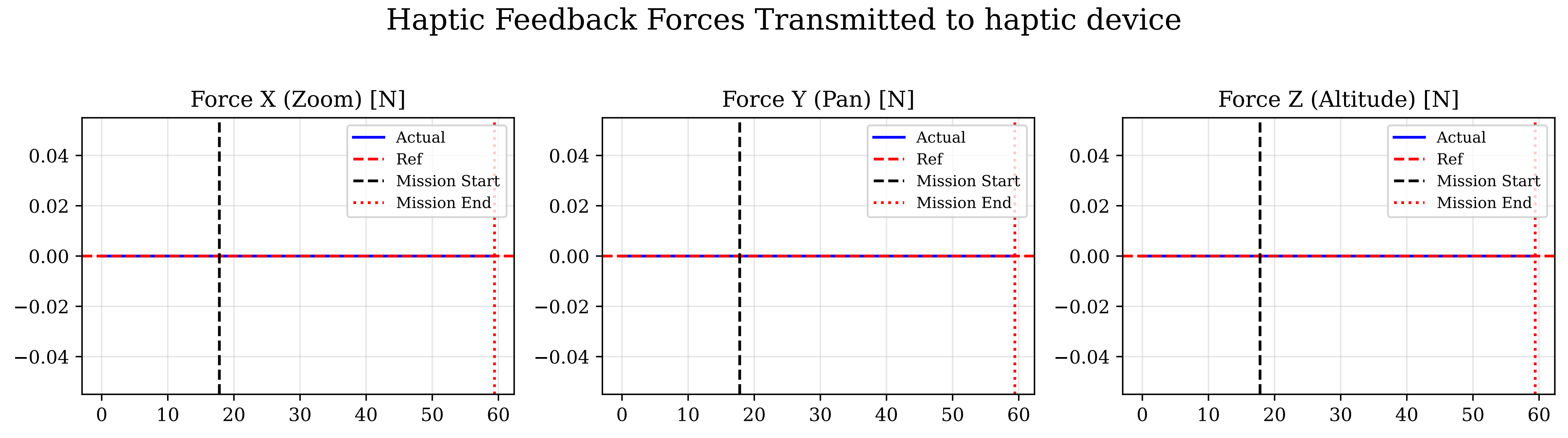
Yaw Tracking: Drone Pointing Toward Target
Yaw Actual vs Desired [rad]
Yaw Error [rad]
Actual
Reference
Mission Start
Mission End
Actual
Ref
Mission Start
Mission End

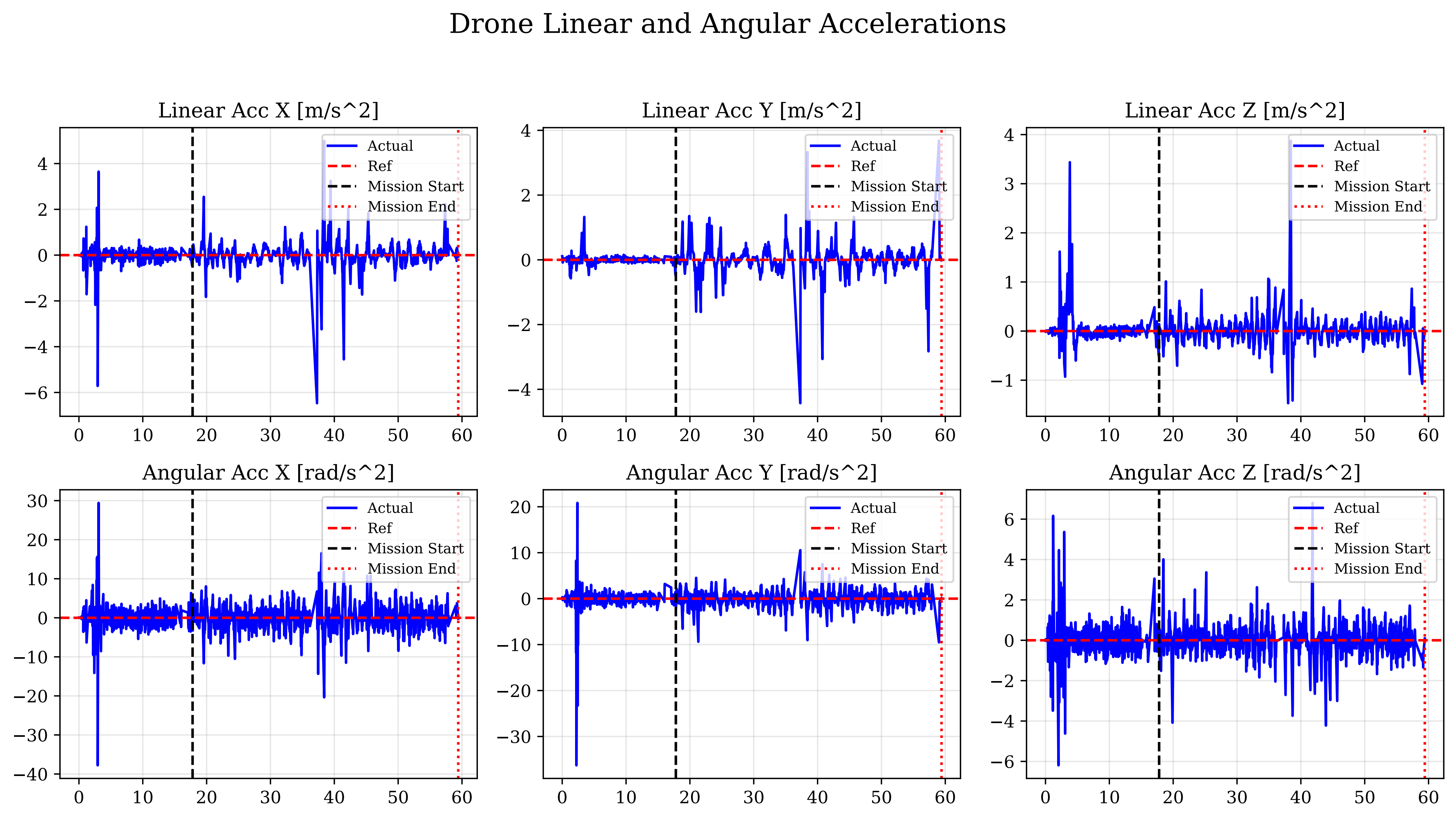

Primary Tracking Errors (Cylindrical)
Cam XY Error [m]
Distance Error |r_cyl_err| [m]
Azimuth Error |beta_err| [rad]
Elevation Error |z_err| [m]
Yaw Error |yaw_err| [rad]
Norm Roll/Pitch Error
Actual
Ref
Mission Start
Mission End


Dynamic States Errors and Feedforward Derivatives
Norm Vel Error [m/s]
Norm Omega Error [rad/s]
Norm Acc [m/s^2]
Norm AngAcc [rad/s^2]
Norm Jerk [m/s^3]
Norm Snap [m/s^4]
Actual
Ref
Mission Start
Mission End

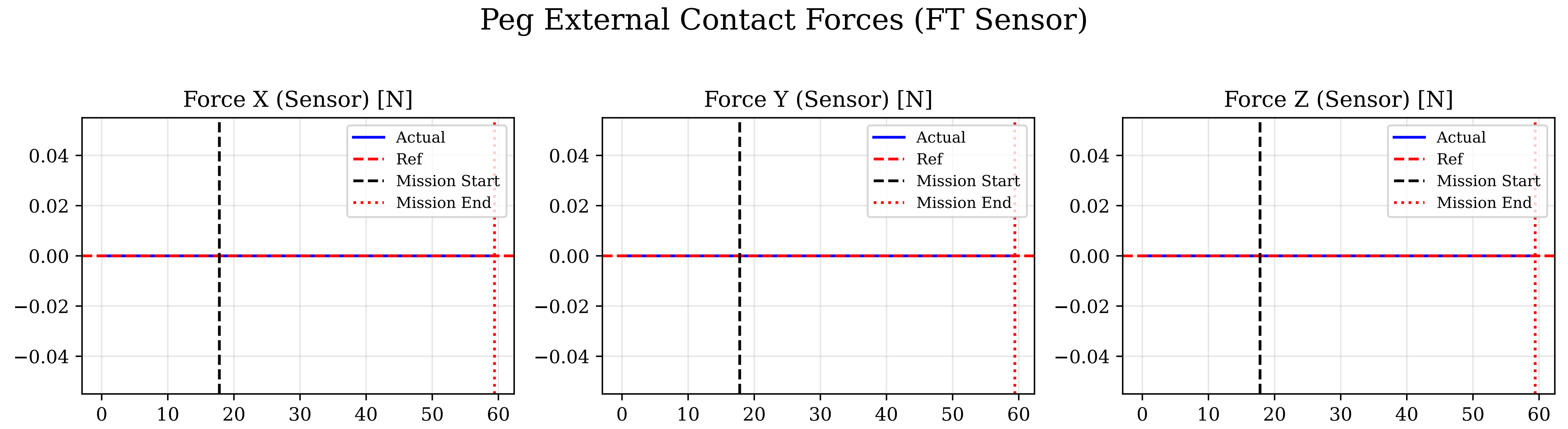


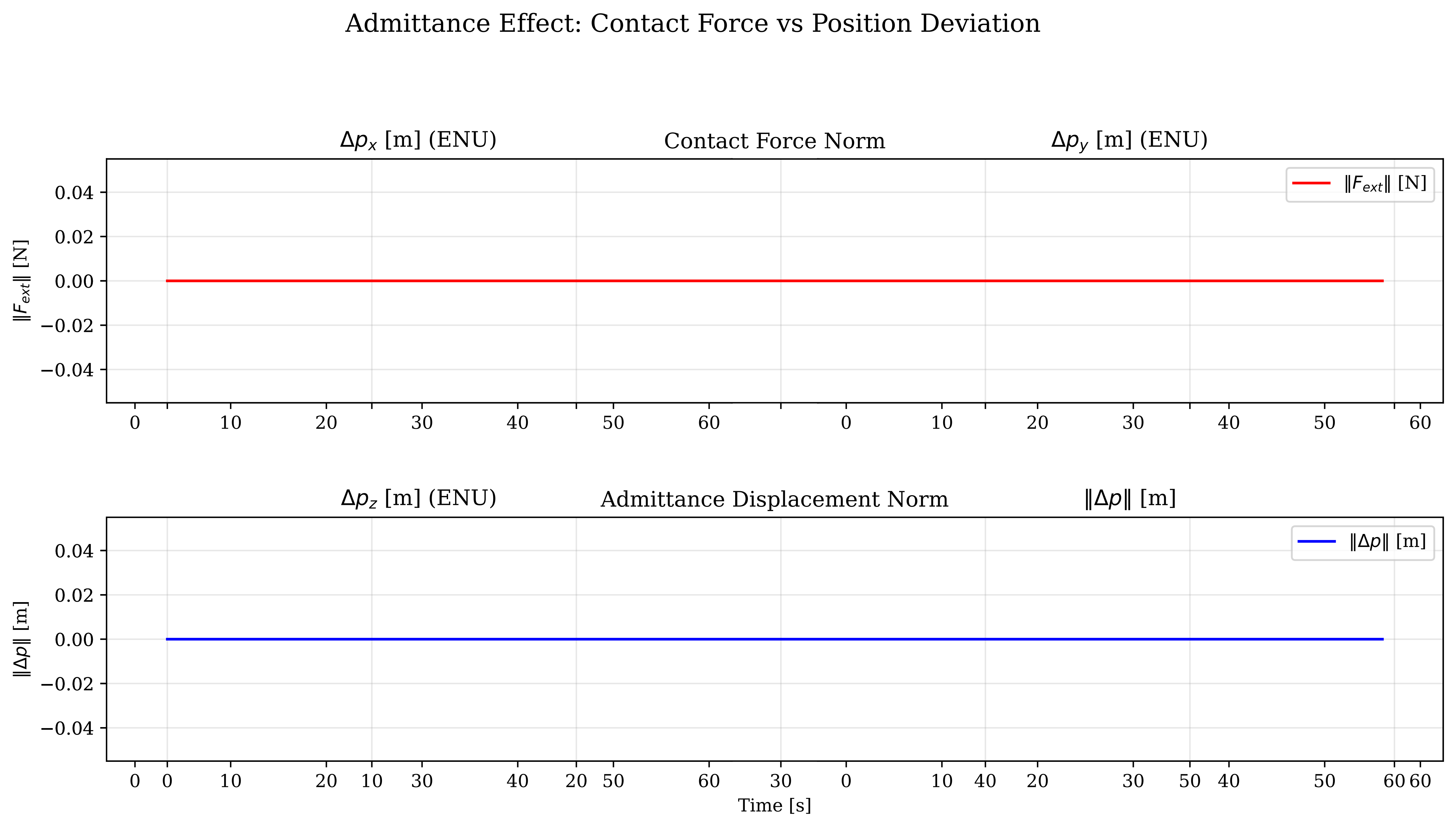
Control Wrench (Hover Force = 20.24N)
Force Z (Thrust) [N]
Torque X [Nm]
Torque Y [Nm]
Torque Z [Nm]
Actual
Reference
Mission Start
Mission End


Haptic Feedback Forces Transmitted to haptic device
Force X (Zoom) [N]
Force Y (Pan) [N]
Force Z (Altitude) [N]
Actual
Ref
Mission Start
Mission End
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
50
60

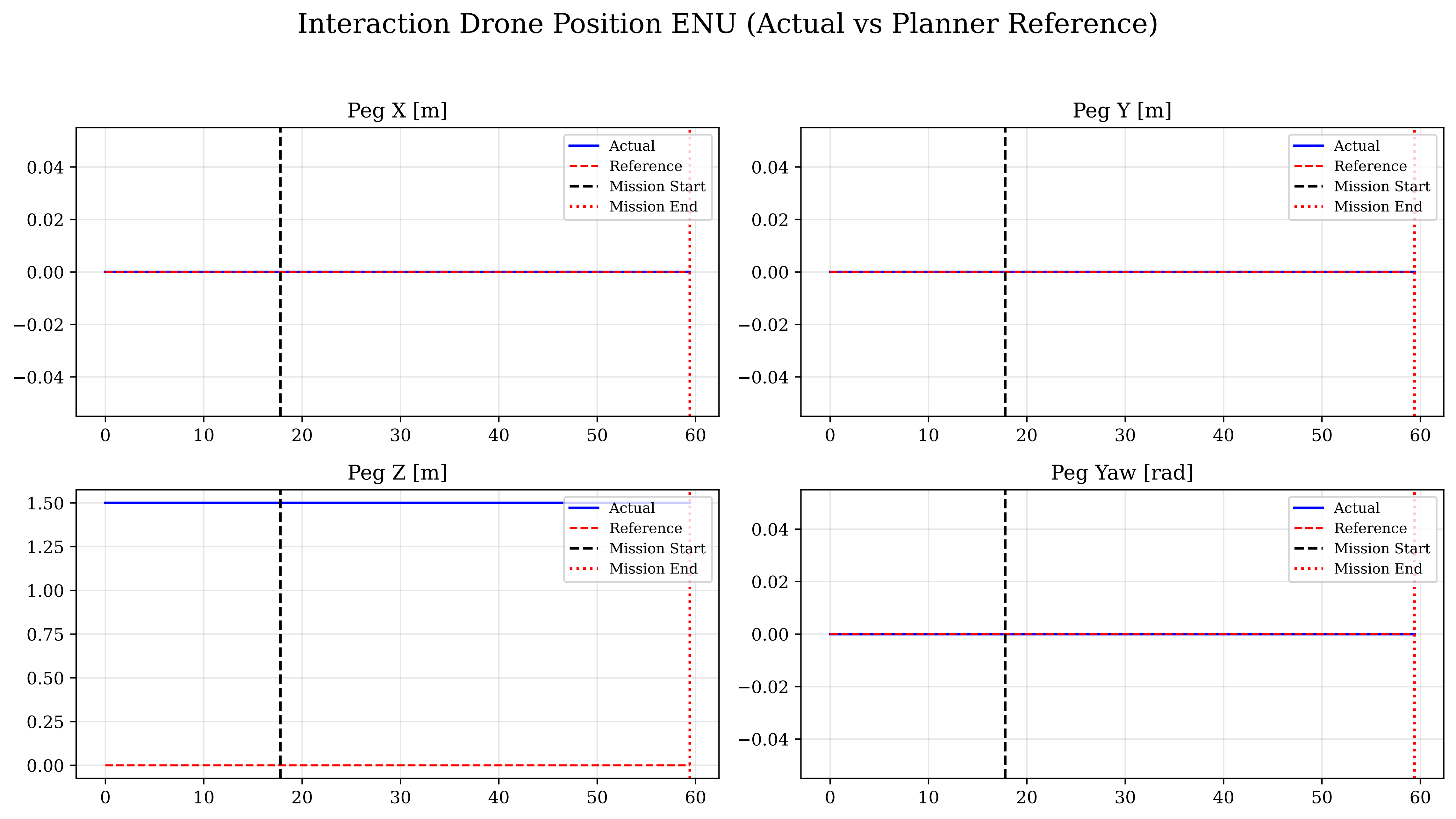


Drone Linear and Angular Accelerations
Linear Acc X [m/s^2]
Linear Acc Y [m/s^2]
Linear Acc Z [m/s^2]
Angular Acc X [rad/s^2]
Angular Acc Y [rad/s^2]
Angular Acc Z [rad/s^2]
Actual
Ref
Mission Start
Mission End

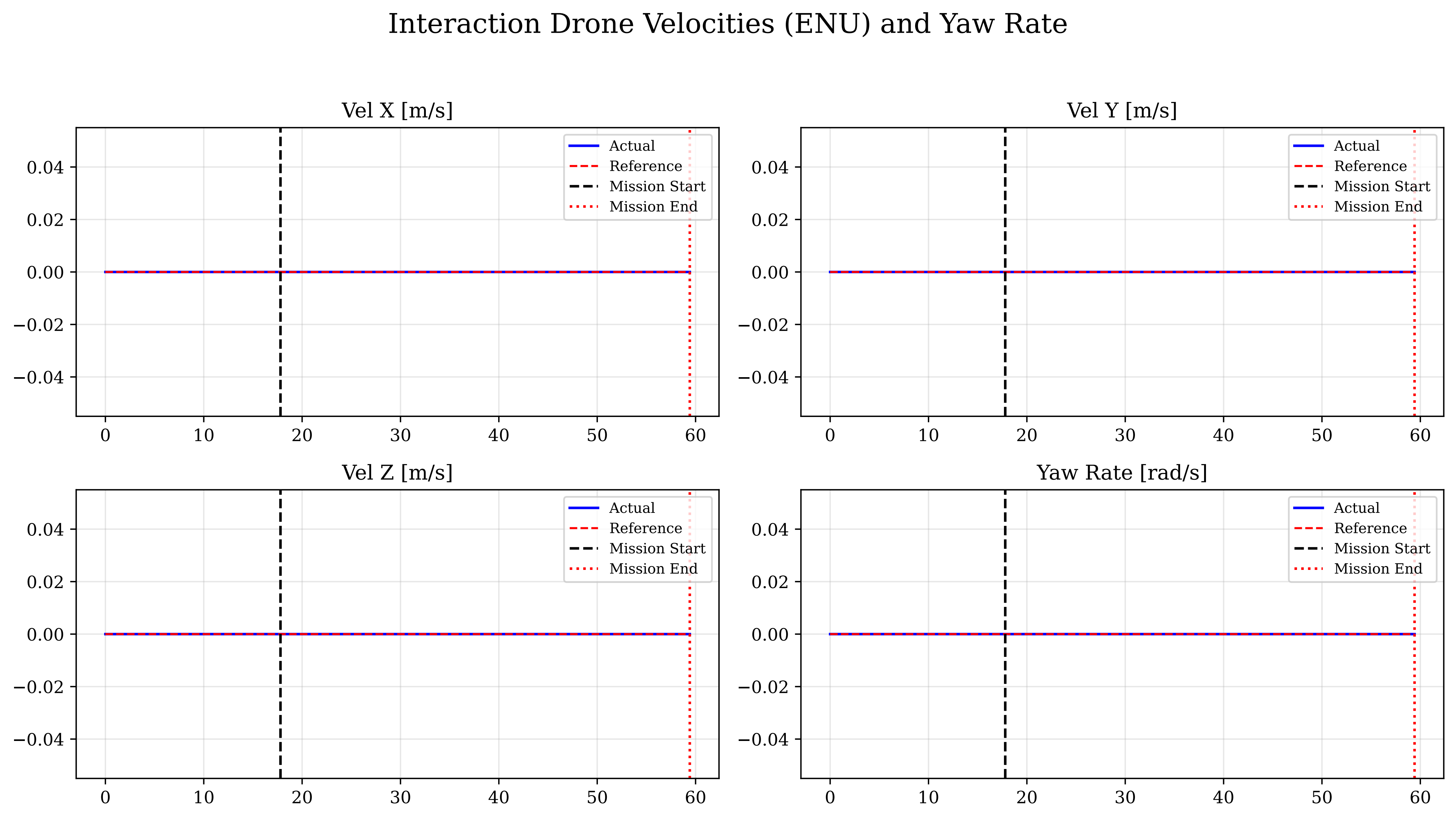

Peg External Contact Forces (FT Sensor)
Force X (Sensor) [N]
Force Y (Sensor) [N]
Force Z (Sensor) [N]
Actual
Ref
Mission Start
Mission End
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
50
60

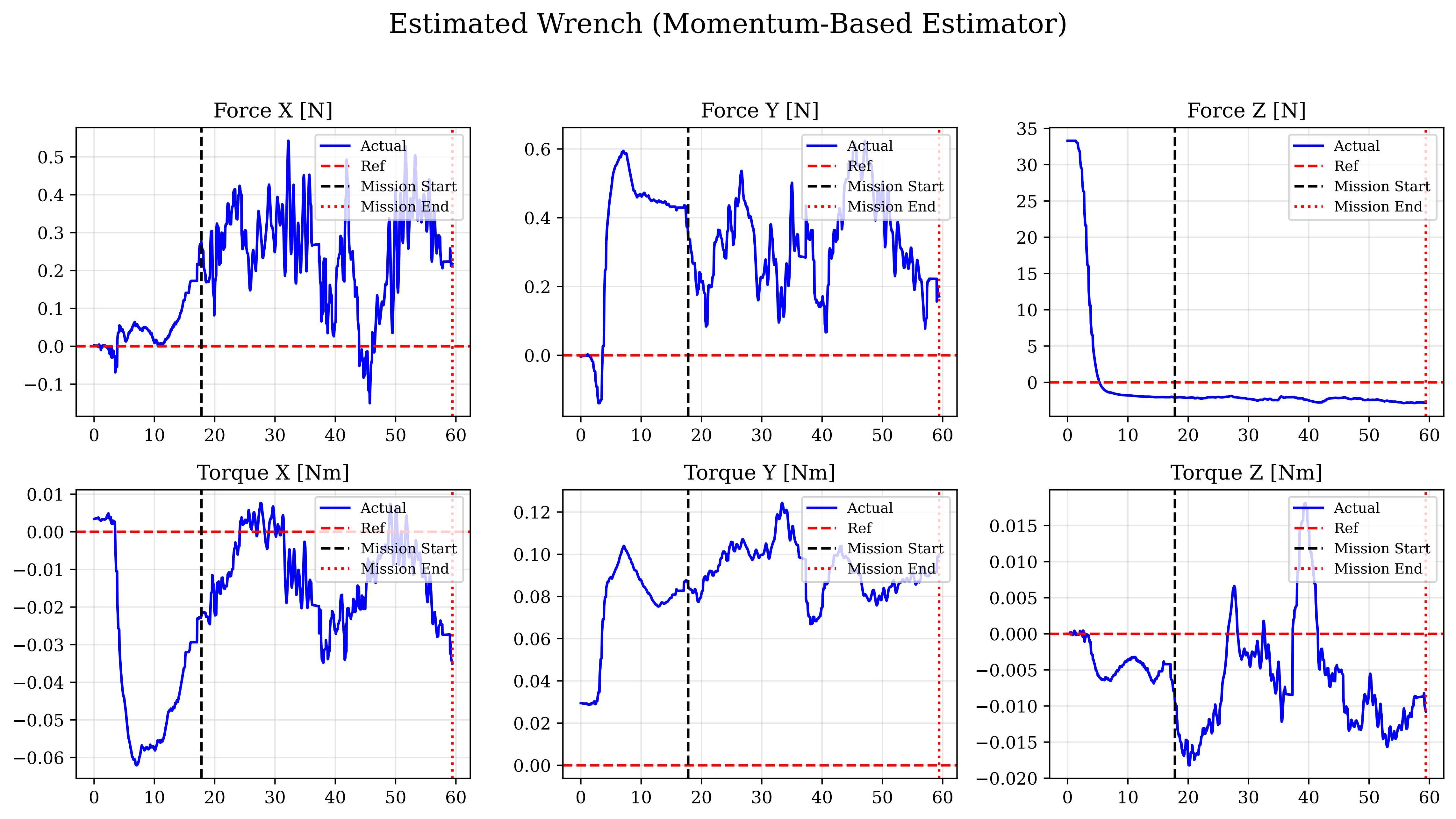


Admittance Effect: Contact Force vs Position Deviation
Δp_x [m] (ENU)
Contact Force Norm
Δp_y [m] (ENU)
$\|F_{ext}\|$ [N]
Δp_z [m] (ENU)
Admittance Displacement Norm
$\|\Delta p\|$ [m]
Time [s]

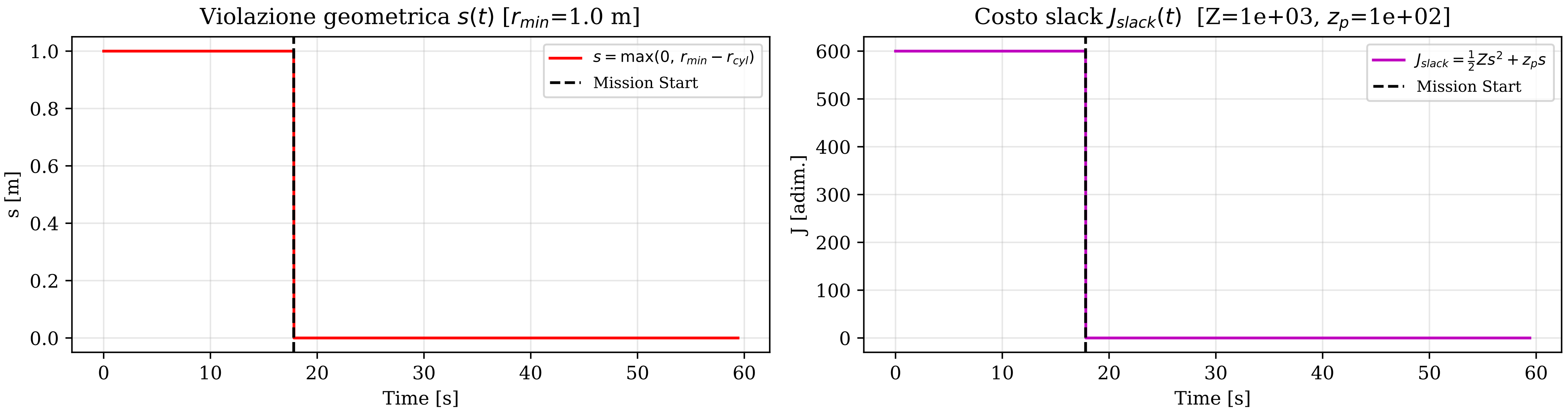


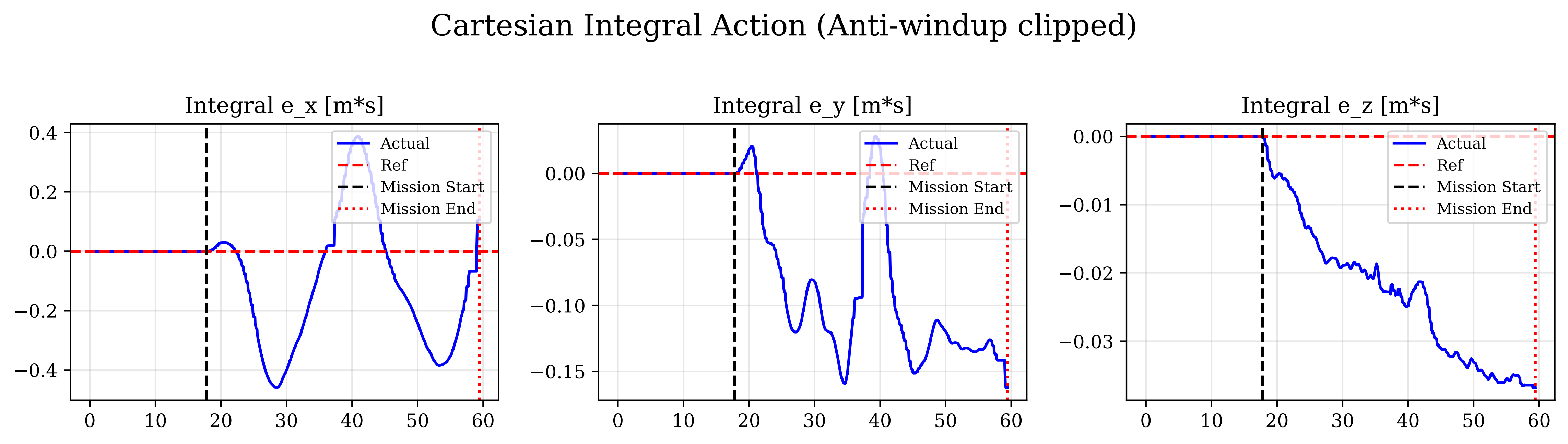
Interaction Drone Position ENU (Actual vs Planner Reference)
Peg X [m]
Peg Y [m]
Peg Z [m]
Peg Yaw [rad]
Actual
Reference
Mission Start
Mission End

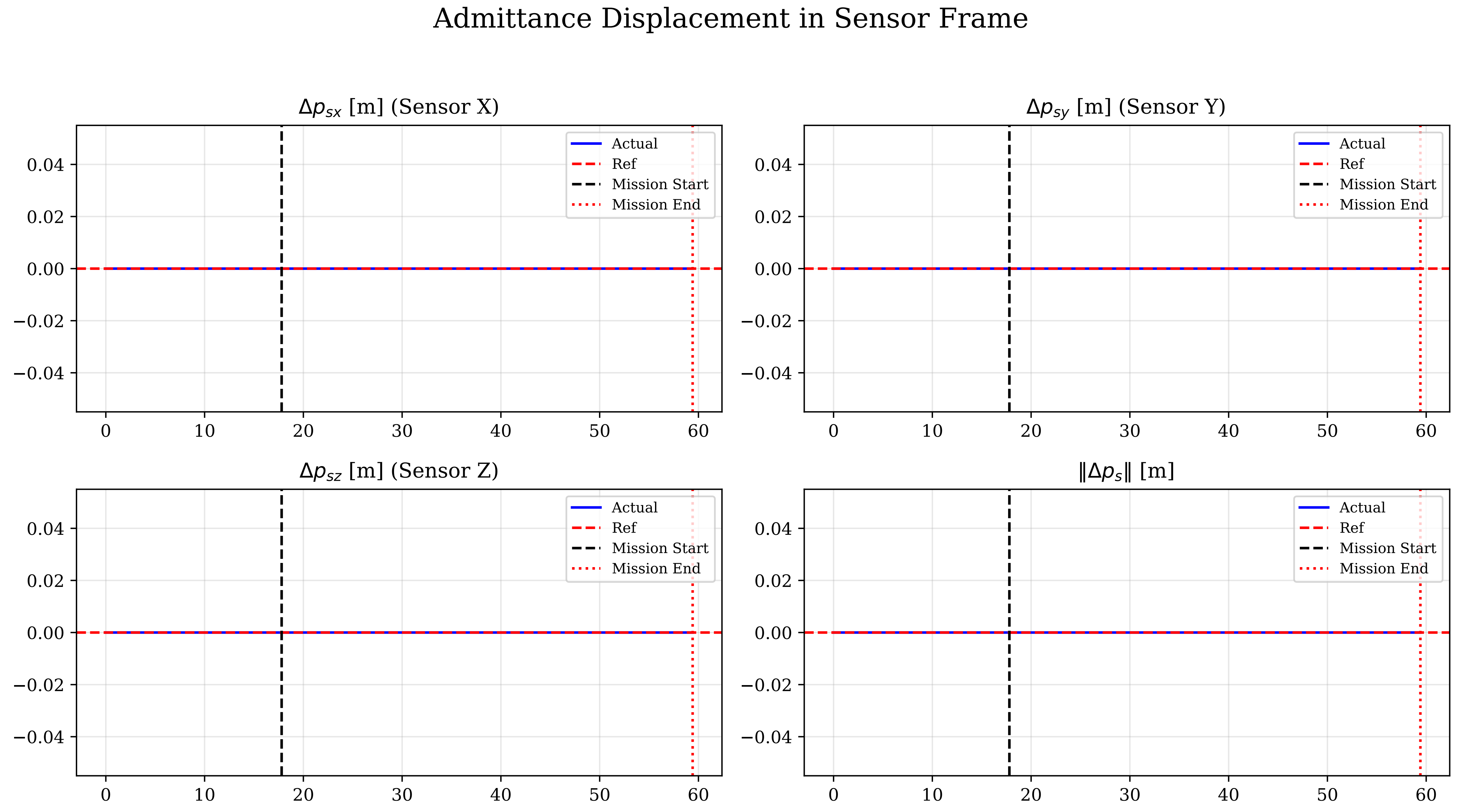

Interaction Drone Velocities (ENU) and Yaw Rate
Vel X [m/s]
Vel Y [m/s]
Vel Z [m/s]
Yaw Rate [rad/s]
Actual
Reference
Mission Start
Mission End
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
50
60


Estimated Wrench (Momentum-Based Estimator)
Force X [N]
Force Y [N]
Force Z [N]
Torque X [Nm]
Torque Y [Nm]
Torque Z [Nm]
Actual
Ref
Mission Start
Mission End

Soft Constraint — Distanza Minima da Oggetto




Cartesian Integral Action (Anti-windup clipped)
Integral e_x [m*s]
Integral e_y [m*s]
Integral e_z [m*s]
Actual
Ref
Mission Start
Mission End
0.4
0.2
0.0
−0.2
−0.4
0.00
−0.05
−0.10
−0.15
−0.01
−0.02
−0.03
0
10
20
30
40
50
60



Admittance Displacement in Sensor Frame
Δp_{sx} [m] (Sensor X)
Δp_{sy} [m] (Sensor Y)
Δp_{sz} [m] (Sensor Z)
$\|\Delta p_s\|$ [m]
Actual
Ref
Mission Start
Mission End
0.04
0.02
0.00
−0.02
−0.04
0
10
20
30
40
50
60